Does Underlying Structure Predict Quantum Trainability and Dequantization?

Charbel A. Dargham

Abstract

Quantum trainability and dequantization has, till this day, been shown post-mortem of the construction and execution of a quantum machine learning algorithm, after alot of effort and resources were put into making it. we wish to find a way to know apriori whether or not a machine learning problem is, first trainable with quantum circuits, and second know if a classical algorithm would beat it. We narrow our vision in this study to a subcase of learning problems.

1. THE QUESTION

The Main Question of This Endeavor. *Can the underlying structure of a learning problem predict, before training, whether structure-preserving quantum models of it will be trainable and resist dequantization?*

1.1 Why Underlying Structure

McClean et al. 2018 (Nat. Commun. 9, 4812) found **barren plateaus**: in random, structure-free parameterised circuits, gradients shrink exponentially with the number of qubits. The training landscape is flat, so the optimiser has no direction to follow.

Building the problem's symmetry into the circuit restricts what the circuit can explore, and the gradients stay large, we site the following. Larocca et al. 2022, *Group-invariant QML*, PRX Quantum 3, 030341: how symmetry enters quantum models. Ragone et al. 2024, Nat. Commun. 15, 7172, with Fontana et al. 2024 as an independent derivation: an exact formula for gradient size in terms of the **dynamical Lie algebra (DLA)**. The DLA is the space of all operations the circuit's generators can produce. A small DLA gives large gradients; an exponentially large DLA gives vanishing gradients. Cerezo et al. 2021, Nat. Commun. 12, 1791: observables that act on only a few qubits keep gradients alive in shallow circuits. This is where locality comes in.

Most notably Cerezo et al. (Nat. Commun. 16, 7907, 2025) argue that the same small DLA that makes a model trainable also lets a classical computer imitate it. That imitation is **dequantization**, and it means the model gives no quantum advantage. Goh et al., $\mathfrak{g}$-*sim*, arXiv:2308.01432, is the concrete classical simulation method.

We decided to expand on this work and perform an experiment a priori of any proving theorems to understand how the interplay between four decided structural features of a learning problem and our desired prediction behaves. We define the four kinds of structure: **symmetry**, transformations that don't change the answer; **locality**, which parts of the input act on each other directly; **spectrum**, which patterns or modes of the input carry the answer; **data complexity**, how hard the inputs are to describe classically.

Only symmetry has a theorem that runs from the problem to the trainability formula. Ragone's formula takes three inputs: the DLA $\mathfrak{g}$, the input state ρ, and the measured observable O. For locality, spectrum and data complexity, that mapping is the project's own working hypothesis, not a published result.

1.2 The Plan

Phase A: Build the dataset of problems, circuits and measurements.

(i) Collect problems: these are real learning problems, we'll discuss this step in Section 2, we have harvested, and after cleaning the data 1368 problems were harvested categorized into 4 types (more about that in Section 2). (ii) Build circuits from each problem. There are three kinds: a **natural model**, built from the problem's structure, for example by **twirling**: averaging each gate over the symmetry group so the gate becomes symmetric; a **generic control** that ignores the structure; a **path model** with a dial λ running from generic (0) to fully structured (1). The controls are there to show whether the structure *causes* good behaviour or only comes with it. (iii) **Fingerprint each circuit** from its description alone, with no training. The main quantities are the size of the space the circuit can reach (the DLA) and how that space splits into blocks. (iv) **Commit predictions before measuring.** Apply Ragone et al. 2024 for trainability and Goh et al. 2023 for the cost of classical simulation, and commit the predictions to git. The commit timestamp proves they came first. (v) **Measure.** Record how much the output varies at random parameters, what happens in training, and what classical simulation cost.

Phase B: build the predictor.

This is a statistical model that goes from fingerprint to (trainable?, hard to simulate?). **Testing:** it is always tested on **problem families it has never seen**, so that it cannot succeed by memorising a family. **Success:** the predictor counts as a success if it beats the theoretical formulas where they don't apply, such as shallow circuits, approximate symmetry, or cases outside the theorems' conditions.

2. PHASE A

2.1 Data Harvest

2.1.1 What does the dataset contain?

At each row of our dataset, we have a problem, one binary learning task at a fixed n: its samples, its TRAIN/TEST split, its label rule, its equation and $\bar{M}$. Effectively, one problem is

$$\mathscr{P}_n = (\mathscr{X}_n,\ \mathscr{Y}_n,\ \mu_n,\ \mathscr{E}_n,\ f^*),$$

whose parts are given in Table 1.

Table 1: The parts of a problem record.

Part	Meaning
$\mathscr{X}_n$	the inputs, on n sites, with n = number of qubits, from 4 to 12
$\mathscr{Y}_n$	the labels, always ± 1
μ_n	the data distribution: the actual samples and their fixed 70/30 TRAIN/TEST split
$\mathscr{E}_n$	the differential equation each input determines, written as one operator $M(x)$ per input
f^*	the target, plus the sentence saying how the labels were made

$\bar{M}$, the task operator. It is built from TRAIN only. $M(x)$ feeds the circuit's encoding; $\bar{M}$ feeds its trainable generators. **Provenance:** source, licence, citation, file hash, group, and the "modelling map" (how a domain object was turned into $M(x)$). **An origin tier:** tier 1, REAL: real inputs and real labels; tier 2, REAL_INPUT_COMPUTED_LABEL: real inputs, labels computed exactly from them; tier 3, SYNTHETIC: both generated.

The type of a problem is fixed by its operator, not by its domain (HARVEST.md §3). The same domain can hide different mathematics: MaxCut encodes like a graph, while 3-SAT has three-qubit terms that a graph does not have (Table 2).

Table 2: The four problem types.

Type	$M(x)$	Equation	Do sites have an identity?
GRAPH	Laplacian $L = B^\top W B$	heat, $\dot{x} = -Mx$	no
COVARIANCE	precision $Q = R^{-1}$, the inverse of the channel correlation matrix	Ornstein–Uhlenbeck, $dx = -Mx\,dt + \text{noise}$	yes (channel C3 is C3 in every trial)
DIAGONAL	clause-penalty Hamiltonian, Z terms only	Schrödinger, with $e^{-i\gamma H}$ as the encoding	no
QUANTUM	Hamiltonian with non-commuting terms; the input is its ground state	Schrödinger, $\dot{\psi} = -iM\psi$	yes (chain sites)

Variables. Each problem is described by two groups of variables (Table 3). The first is recorded at harvest time, one row per problem in `data_final/harvest.sqlite`; the second is computed from the factor graph of each input $M(x)$, a bipartite graph with one node per site and one per term of the operator, and summarised over the problem's inputs by the median and the interquartile range. All twelve structure variables are unchanged when the operator is multiplied by a constant.

Equation checks (HARVEST.md §4). These are run on every input. Failures are counted, never fixed, and the final count is 0 failures. Graphs: L is symmetric with zero row sums, and the number of zero eigenvalues equals the number of connected components. Covariance: $QR = I$. QUANTUM: $H\psi = E\psi$, and E is the lowest eigenvalue. SAT: the Pauli expansion equals the violated-clause count on all 2^n assignments.

The audit script ran on the raw data and found **1 379 problems and 0 structural failures**. Files, counts, splits, labels, equation checks, finite inputs, provenance and unique ids were all fine. The problems were in the content (Table 4).

Raw data were never modified. The final dataset was built from the raw harvest by a script with six stated rules (R1–R6), and the raw data's sha256 is identical before and after the build.

Table 3: Variables of the dataset.

Variable	Meaning
Recorded per problem	
type, n	operator type (Table 2) and number of sites (qubits), 4 to 12
samples, TRAIN/TEST	number of labelled inputs and their fixed 70/30 split
class balance	share of $+1$ labels, kept within $[0.2, 0.8]$
label rule	how the labels were made (e.g. property of the graph, score above the TRAIN median, side of a phase transition)
distinct inputs, conflict rate	number of different inputs, and share of equal inputs carrying different labels
source, group, tier	where the problem comes from, its independent source group (58 in all), and its origin tier
equation check	check method, worst residual and number of failures
Structure, computed from the factor graph of each input	
terms per site	number of terms divided by n: how many interactions per site
term sizes (mean; fractions of size 1, 2, ≥ 3)	how many sites each term acts on: pairwise or many-body
site degree CV	spread of the number of terms per site over its mean: 0 when every site is equally involved
support density	share of site pairs that share at least one term: how far interactions reach
symmetry classes, symmetry fraction	classes of sites that colour refinement cannot tell apart; 1 when all sites look alike, 0 with no symmetry
spectral entropy	entropy of the normalised Laplacian eigenvalues of the factor graph: low when a few modes dominate
participation ratio	$(\sum\lambda)^2/\sum\lambda^2$ over the number of nodes: effective share of modes in use
algebraic connectivity	second-smallest Laplacian eigenvalue: how hard the graph is to cut in two

Problems found in the raw harvest (1 379 problems, 0 structural failures): (1) **Leak:** in 22 problems, identical inputs sat on both sides of TRAIN/TEST. (2) **Class balance:** 1 problem was below the 0.2 floor. (3) **Degenerate ground states:** 24% of spin-chain samples, so the stored state was arbitrary. (4) **Unreliable labels:** 2 molecule geometries where Hartree–Fock did not converge. (5) **Uninformative operators:** 475 sensor samples whose operator is the identity matrix, which acts as a label shortcut.

What was done: Spin chains re-harvested with ground states computed inside a fixed symmetry sector (fixes 3). Samples with gap $< 10^{-6}$ fell from 642 to 10 on a dry run. **R1:** excluded the unbalanced problem (fixes 2). **R2:** group-aware re-split, keeping identical inputs on one side, then recomputing labels and $\bar{M}$ on the new TRAIN (fixes 1). **R3:** weights so each distinct input counts once in TEST. **R4:** at least 30 distinct TEST inputs. **R5:** removed the non-converged samples (fixes 4). **R6:** flagged rather than removed the 475 uninformative and 26 degenerate samples (handles 5). A sensitivity check runs with and without them.

Final state: 1 368 problems, 58 groups, 337 930 samples. Audit: 0 FAIL. TRAIN/TEST leaks went from 22 problems to 0. **2 WARN:** two TEST balances at 0.17, handled by reporting balanced accuracy. **One known gap:** XY-periodic (10 problems) is missing because a solver failure made the harvest reject the whole dataset. The bug is fixed, but the re-harvest is still pending.

2.1.2 Do the four types differ in structure?

Take one structure feature at a time. Once size n and source are accounted for, does its value depend on the type?

It tests one way the "extract structure" step could fail. failure as features that are "identical everywhere, or different only by construction". There are two possibilities. **Features identical across types** carry no information, and could never predict anything about circuits. **Features that differ only because of how a type is defined** are not evidence of anything. For example, every COVARIANCE precision matrix is dense, so its support density is ≈ 1 by definition. A test that "finds" this has found nothing.

So Q2 asks whether the extracted structure varies *beyond what we built into it.* we decided to fit each feature into a hierarchicha generalized addtitive mixed model (Wood 2017; Pedersen et al. 2019)

$$\text{feature} = \text{type baseline} + \text{smooth curve in } n \text{ for each type} + \text{offset for each group} + \text{noise}.$$

a B-spline, i.e. short polynomial pieces joined smoothly, limited to 3 degrees of freedom. That is enough to follow a curve like $2/n$ but not enough to chase noise. Each source gets its own baseline, drawn from a common distribution. This handles the fact that problems from one source are not independent. Treating them as independent can push the false-positive rate from 0.05 up to 0.80 (Aarts et al. 2014).

Table 4: Findings of the audit of the raw harvest.

#	Finding	Size	Cause
1	**Leak:** identical inputs on both sides of TRAIN/TEST	22 problems (42 contain duplicates at all). Worst: NAE-3 $n = 4$, where 71 of 90 TEST inputs are also in TRAIN	random formulas repeat at small n; mirror-image molecules give the same graph; EigenWorms stores repeated recordings
2	**Class balance below the floor**	1 problem: XOR-3 $n = 10$ count-high, overall 0.19, TEST 0.14	the median-tie rule only checked TRAIN
3	**Degenerate ground states:** the stored vector is an arbitrary point of the ground space	10 005 of 41 920 spin samples (24%), in 127 of 262 problems	physical: odd-n chains that conserve total spin, ordered phases, cluster-Ising edge states
4	**Hartree–Fock did not converge**, so the labels are unreliable	2 H_2O geometries, each present in 2 problems	numerical
5	**Uninformative operator:** Ledoit–Wolf shrinkage ≥ 0.99, so the operator is the identity matrix	475 UEA samples (LSST 183, RacketSports 292)	30–36 time points with bursty signals

We effectively can read the following off the model. **ICC (intraclass correlation)** = between-group variance / total variance. Converted into an **effective sample size** (Kish): how many independent observations the feature really has. **Level test:** do the types differ at equal n? This is a likelihood-ratio test of the model with type terms against the model without them. **Shape test:** do the types scale differently with n? One curve per type against one shared curve. **Holm correction** over the 12 features. It keeps the chance of even one false "difference" at 5%. **Marginal** R^2 (Nakagawa & Schielzeth 2013): the share of variance explained by type and n alone, without the group offsets. **Pairwise type differences** at $n = 8$, 10 and 12.

Every forced value matches its derivation exactly (`q2_pipeline_check` is empty). One bug was found this way and fixed before any result was read: factor labels depended on site order.

The spectral features are the clear finding. Spectral entropy, participation ratio and algebraic connectivity: **level:** they differ by type, with p from 10^{-12} to $5 \cdot 10^{-7}$; **shape:** for entropy and participation ratio, the types also scale differently with n (shape $p \approx 10^{-23}$, $2 \cdot 10^{-28}$); **effect size:** marginal R^2 of 0.30–0.55. **GRAPH stands apart.** It differs from every other type on all three spectral features (pairwise $p < 10^{-4}$). COVARIANCE, DIAGONAL and QUANTUM are mostly indistinguishable from each other (spectral entropy at $n = 10$: 0.973, 0.963, 0.979; pairwise p 0.21–0.66). **Locality features** (support density, site-degree spread, terms per site, symmetry classes) differ in level but not in shape. Their "QUANTUM" finding rows are only the 6 molecules at $n = 8$ and 12, because the spin-chain cells are forced. **Term-size fractions and the symmetry fraction show no difference** after Holm correction (p 0.08–1). **ICC is 0.5–0.9**, so the effective sample size is **18–132, not 1 107.** The group-level model was necessary. **Sensitivity:** results are identical with and without the flagged samples.

The implications below are from ANALYSIS_RESULTS "What the study supports" §2 and §5. (1) **The extracted structure varies beyond construction**, most robustly through the weighted spectral features. It is not a relabelling of the definitions. (2) **The variation is coarse.** It separates graphs from everything else, but not covariance, SAT and quantum Hamiltonians from each other. A predictor built on these features will see roughly two kinds of operator, not four. (3) **Sources dominate.** Because the effective sample size is small, the circuit phase should favour diversity of sources over more problems per source, and should validate by holding out whole sources.

2.1.3 Can a problem's kind be recovered from its structure alone?

if the features only re-label the type, they carry nothing beyond the definitions. We performed a logistic regression from the features to the class. to valitdate we trained on all sources but one, test on the held-out one and output a score for balanced accuracy. **Three size controls:** an n-only classifier as a baseline; the Dinga et al. 2020 test, which asks whether the predictions add information beyond n; a subset balanced for size. **Two layers: Q3a** recovers the 4 types. This is a pipeline check. **Q3b** recovers the domain within a type. This is the finding.

as for the results: **Q3a: 0.91** (chance 0.25, n-only 0.39). The errors fall where the operator definitions predict. For example, NAE-3 and 2-SAT have only pairwise terms and are classed as GRAPH. **Q3b is negative.** COVARIANCE scores 0.49 (chance 0.50). GRAPH (0.08) and DIAGONAL (0.20) are below chance, a known effect of leave-one-group-out with few sources (Jamalabadi et al. 2016). QUANTUM's 1.00 is trivial: forced spin-chain features separate it.

the features identify the kind of operator, not the domain within a type. There are two open readings: the features are too coarse, because the factor graph loses how terms overlap and the Hodge complex would keep it; or the domain is not the partition that matters for trainability.

2.1.4 How does structure change with size n?

A predictor trained at $n \leq 12$ must extrapolate, and a feature with no regular dependence on n cannot be extrapolated. Each sub-family is followed across n as one line (43 of 64 lines have enough sizes), and each feature is read in three tiers: the slope against $\log n$ with its 95% interval; a test of laws known in advance, exact for forced laws and by likelihood ratio for predictable ones; and, where no law is known, a choice among constant, $a + b/n$, $a + b\log n$, power and exponential, named only if $\Delta\text{BIC} > 10$ (Kass & Raftery 1995). On the spin chains, two physics laws are tested on each side of every transition: ground-state entanglement grows as $(c/6)\log n$ on open chains and $(c/3)\log n$ on periodic ones at criticality (Calabrese & Cardy 2004) and stays bounded where the chain is gapped (Hastings 2007); the in-sector gap falls as $1/n$ at criticality.

The structure changes monotonically almost everywhere: 379 of 428 trends are monotone, spectral entropy rises on 41 of 43 lines and support density falls on all 39. All 28 forced laws match exactly. Among the predictable SAT laws, PLANTED-3 agrees with its prediction, while NAE-3 deviates in a way explained by its pairwise terms cancelling. Only 19 of 177 laws can be named, mostly on the EEG lines; the other 158 are undetermined on $n = 4$–12. On the open XXZ chains in their critical phase, the entanglement slope is 0.18–0.21 against the predicted $c/6 = 0.167$, inside the interval in all four cases. Four readings become flat when the flagged degenerate ground states are removed, and are reported as depending on them.

The extracted structure therefore changes lawfully with n and reproduces known physics, but most laws cannot be identified at $n \leq 12$: a predictor trained on these data describes finite sizes, not asymptotics.

3. CONCLUSIONS

The study validates the second link of the chain, the extracted structure, before the expensive steps that follow; it says nothing directly about trainability or dequantization, which require simulated circuits. Within that scope, the extraction is correct: every forced value and every forced law matches exactly, the one bug found by these checks was fixed before any result was read, and the type-classification errors fall where the operator definitions predict. The structure is varied beyond construction, most robustly through the weighted spectral features, though coarsely: GRAPH separates from the other three types, while COVARIANCE, DIAGONAL and QUANTUM remain close. It changes lawfully with n, with the critical XXZ chains following the Calabrese–Cardy law. It identifies the kind of operator but not the domain within a type. Finally, the data hold far fewer independent observations than problems (ICC 0.5–0.9, effective sample size 18–132), so the circuit phase should favour diversity of sources and validate the predictor by holding out whole sources.

4. NEXT STEPS

The permutation tests of Q2 (5 000 group-level shuffles per feature) and Q3 (1 000 per task) remain to be run; until then the p-values rest on the mixed models' assumptions. The robustness study (Q6) was not carried out, so robustness currently rests on the R6 sensitivity runs and the size controls of Q3. The analysis choices left open (classifier regularisation, the correlation-block threshold, the minimum number of sizes per line, log transforms) must be confirmed. On the data side, the XY-periodic spin chains are to be re-harvested with the corrected solver, and the origin tier of the molecule problems aligned across files. On the structure side, Hodge features built on the factor graph would keep how terms overlap, which the current features lose, and are the first candidate to test the "features too coarse" reading of Q3b. The project then moves to the rest of Phase A: building natural, generic and path circuits from each problem, fingerprinting them, committing the theoretical predictions, and measuring trainability and the cost of classical simulation.